Assignment – 1.17

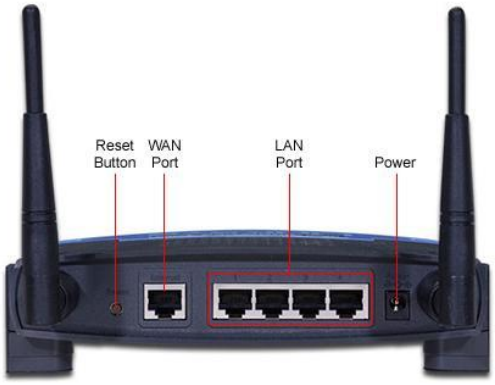
Introduction to Network Devices Theory

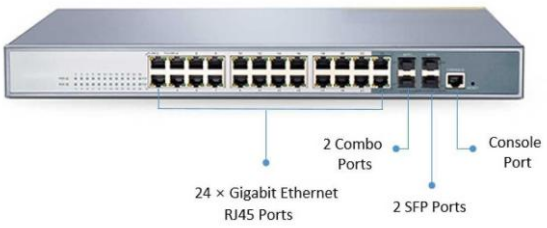
Q - 1. Create a table listing the main functions and typical ports used by switches, routers, hubs, firewalls, and access points in a home WiFi setup.

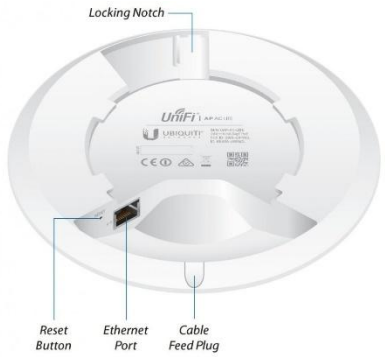
- **Main functions and ports of network devices:**

Device	Main function	Common ports
Switch	Connects many wired devices.	Ethernet, console, power
Router	Connects a network to the internet.	WAN, LAN, power
Hub	Sends data to all connected devices.	Ethernet, power
Firewall	Protects the network from unwanted traffic.	Ethernet, console, power
Access Point	Provides Wi-Fi to phones and laptops.	Ethernet, power, antenna

<u>Q - 2.</u> Given images or diagrams of network devices, label each device and mark the ports (Ethernet, console, power, antenna) for at least three devices.







Q - 3. Pick a popular app you use daily (like Instagram, Zomato, or Paytm) and describe in 3-4 lines how at least two different network devices (from switches, routers, firewalls, access points) play a role when you use that app.

- **How network devices work with Instagram:**

  When I use Instagram, my phone connects to an **access point** using Wi-Fi.

  The **router** connects my home network to the internet.

  The **firewall** helps block unwanted network traffic. These devices help Instagram work properly.

Q - 4. Suppose your college WiFi is slow and sometimes disconnects. Based on the features of access points, switches, and routers, suggest one possible reason for the issue and how upgrading one device could help. Hint: Think about device capacity, coverage, and traffic handling.

- **College Wi-Fi is slow and disconnects:**

  The college Wi-Fi may be slow because too many students are connected to the same access point.

- **Solution:**

  Upgrade the access points or add more access points. This improves Wi-Fi coverage and allows more students to connect smoothly.